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In [1]: import pandas as pd
import numpy as np
```

```
In [2]: #Task 1: Loads CSV and displays first 5 rows.
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```
ipl_df = pd.read_csv("IPL_matches_assignment.csv")
print(ipl_df.head(5))
```

	id	season	team1	team2	\
0	1	2020	Mumbai Indians	Chennai Super Kings	
1	2	2020	Chennai Super Kings	Royal Challengers Bangalore	
2	3	2020	Royal Challengers Bangalore	Kolkata Knight Riders	
3	4	2021	Mumbai Indians	Kolkata Knight Riders	
4	5	2021	Chennai Super Kings	Royal Challengers Bangalore	

	winner	win_by_runs	win_by_wickets
0	Mumbai Indians	5	0
1	Chennai Super Kings	0	7
2	Royal Challengers Bangalore	20	0
3	Mumbai Indians	10	0
4	Chennai Super Kings	0	6

```
In [5]: # TASK 2: Removes missing winner and resets index
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```
ipl_df = ipl_df.dropna(subset=["winner"]).reset_index(drop=True)
```

```
In [6]: #Task 3: Creates match_margin_type.
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```
ipl_df["match_margin_type"] = np.select([ipl_df["win_by_runs"] > 0, ipl_df["win_b
print(ipl_df[["winner", "win_by_runs", "win_by_wickets", "match_margin_type"]])
```

	winner	win_by_runs	win_by_wickets	match_margin_type
0	Mumbai Indians	5	0	Runs
1	Chennai Super Kings	0	7	Wickets
2	Royal Challengers Bangalore	20	0	Runs
3	Mumbai Indians	10	0	Runs
4	Chennai Super Kings	0	6	Wickets
5	Kolkata Knight Riders	7	0	Runs
6	Delhi Capitals	0	8	Wickets
7	Chennai Super Kings	15	0	Runs
8	Royal Challengers Bangalore	0	5	Wickets
9	Mumbai Indians	25	0	Runs
10	Chennai Super Kings	0	8	Wickets
11	Royal Challengers Bangalore	12	0	Runs

```
In [7]: #Task 4: Adds home city for both team1 and team2.
```

```
teams = pd.DataFrame({"team": ["Mumbai Indians", "Chennai Super Kings", "Royal Cha
"Kolkata Knight Riders", "Delhi Capitals", "Gujarat Titans"],
"home_city": ["Mumbai", "Chennai", "Bangalore", "Kolkata", "Delhi", "Ahmedabad"]})
```

```
In [8]: ipl_df = ipl_df.merge(teams.rename(columns={"team": "team1", "home_city": "team1_
on="team1",
how="left")
```

```
In [9]: ipl_df = ipl_df.merge(teams.rename(columns={"team": "team2", "home_city": "team2_
on="team2",
```

```
how="left")
```

```
In [10]: print(ipl_df[["team1", "team1_city", "team2", "team2_city"]])
```

```

      team1 team1_city      team2 \
0      Mumbai Indians      Mumbai      Chennai Super Kings
1      Chennai Super Kings      Chennai      Royal Challengers Bangalore
2      Royal Challengers Bangalore      Bangalore      Kolkata Knight Riders
3      Mumbai Indians      Mumbai      Kolkata Knight Riders
4      Chennai Super Kings      Chennai      Royal Challengers Bangalore
5      Kolkata Knight Riders      Kolkata      Delhi Capitals
6      Mumbai Indians      Mumbai      Delhi Capitals
7      Chennai Super Kings      Chennai      Kolkata Knight Riders
8      Royal Challengers Bangalore      Bangalore      Delhi Capitals
9      Mumbai Indians      Mumbai      Chennai Super Kings
10     Chennai Super Kings      Chennai      Gujarat Titans
11     Royal Challengers Bangalore      Bangalore      Kolkata Knight Riders

      team2_city
0      Chennai
1      Bangalore
2      Kolkata
3      Kolkata
4      Bangalore
5      Delhi
6      Delhi
7      Kolkata
8      Delhi
9      Chennai
10     Ahmedabad
11     Kolkata

```

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In [11]: #Task 5: Creates season-wise wins pivot table and finds the top 3 team-season co

pivot = pd.pivot_table(ipl_df, index="winner", columns="season", values="id", aggfun

print(pivot)

top_3 = pivot.stack().sort_values(ascending=False).head(3)

print("\nTop 3 teams with most wins in a single season:")
print(top_3)
```

```

season      2020  2021  2022  2023
winner
Chennai Super Kings      1    1    1    1
Delhi Capitals            0    0    1    0
Kolkata Knight Riders     0    1    0    0
Mumbai Indians            1    1    0    1
Royal Challengers Bangalore 1    0    1    1

```

Top 3 teams with most wins in a single season:

```

winner      season
Chennai Super Kings  2020      1
                   2021      1
                   2022      1
dtype: int64

```

```
In [ ]:
```